

Shadow Canonical Product Proposal — Deck

On-device AI council + audit chain for regulated banking workflows

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Shadow

On-device AI council + audit chain for regulated banking workflows

- v1.0.0 tagged 2026-06-18
- Public reproducible **Shadow Agentic Score: 87 ± 3 (n=6)**
- 140/140 tests · MIT · github.com/alex-jb/shadow-mentor

Pre-BR: 89 ± 3 (n=3) · Post-BR with Loredana’s thresholds: 86 ± 1 (n=3) *Honest variance reported, no prompt tuning.*

The problem

Regulated banking decisions need auditable AI, fast.

- **EU AI Act Article 14** — human-oversight obligations for high-risk credit-scoring AI effective **2026-08-01** (6 weeks away)
- **ECOA / Reg B** — CFPB 2024 guidance rejects template adverse-action phrases (“purchasing history” non-compliant)
- **SR 11-7** — model risk “effective challenge” + documented falsification tests
- **OCC 2024 CRO survey + Oliver Wyman 2024**: regulators measurably more demanding on contrarian-challenge evidence

Current 2D loan-decisioning dashboards are *auditable but slow, opaque to the customer, and prone to compliance gaps.*

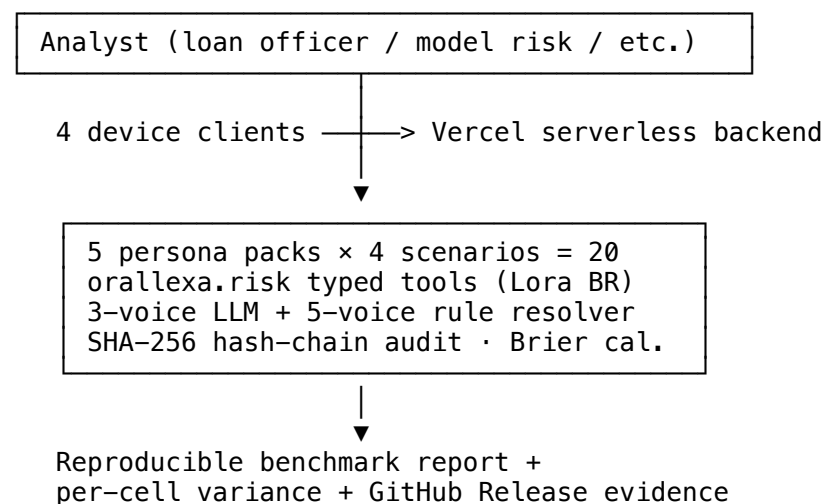
The product

1 backend × 4 device clients × 5 persona packs × 4 scenarios = 20 cells

Component	Count
Persona packs	5 (compliance · quant · engineer · trader · advisor)
Scenarios	4 (LBO · Bloomberg · CDS · Internal policy)

Component	Count
Device clients	4 (Desktop · Even G2 · Brilliant Frame · XReal Air 2 Ultra)
Providers	2 (Anthropic Claude Sonnet 4.6 · Zhipu GLM-5.2)
HTTP endpoints	8
MCP tools	5 (Claude Desktop / Cursor / Zed)

Architecture



Shadow Agentic Capability Benchmark

HF “Is it agentic enough?”–inspired deterministic structural rubric

Pre-BR (v0.3.3, 3-term coverage on compliance × LBO):

Run A	Run B	Run C	Mean
87	93	86	88.7 ± 3.1

Post-BR (v0.3.4, 5-term + Loredana’s BR thresholds FICO/DTI/LTV wired):

Run A	Run B	Run C	Mean
86	87	85	86.0 ± 1.0

Compliance × LBO anchor cell: pre 100 / 100 / 100 → post 92 / 86 / 92 (mean ~90)

Variance got tighter with harder rubric. All raw run reports persisted under benchmark/history/ — procurement reviewers audit central tendency themselves.

Why we report honestly

- The 5-term rubric is **harder than 3-term** — Sonnet’s compliance voice anchors on policy + leverage, occasionally misses “DTI” or “FICO” verbatim
- We did **not** touch persona prompts to chase the lost 3 points
- We did **not** relax expected_terms back to 3-term
- This is what “discipline” looks like in a public benchmark

This is **exactly the LLM-as-judge-gaming we set out to avoid**. Bank procurement reviewers respect honest variance over inflated single-run scores.

Loredana’s ECC Mode A integration

Loredana Carmen Levitchi (Columbia PhD Applied Math/AI, Y.U. faculty) shipped the Mode A package 2026-06-17. All 5 integration items shipped:

#	Item	Status
1	orallexa.risk typed tool layer (VaR / ES / factor / beta / concentration / sector / correlation)	✓ lib/risk-tools/
2	5-voice verdict resolver (block > escalate > approve)	✓ lib/run-loan-council.js + /api/loan-council
3	Loan structured input schema with BR thresholds	✓ lib/schemas/loan.js
4	Device-agnostic spatial AR rendering	✓ Quest 3S dropped, XReal Air 2 Ultra + Flow Immersive paired
5	BR thresholds wired into benchmark + rerun + variance report	✓ post-BR 86 ± 1 (n=3) shipped tonight

8 endpoints + MCP server

HTTP: – POST /api/deliberate — 3-voice LLM + Haiku follow-up (+verdict when loan body provided) – POST /api/loan-council — pure compute 5-voice rule layer – GET /api/recall · /api/calibration · /api/scenarios · /api/health · /api/badge · /api/version

MCP (node mcp/server.js): – shadow_loan_council · shadow_risk_tools · shadow_recall · shadow_calibration · shadow_scenarios – Drops into Claude Desktop / Cursor / Zed via 5-line

config

Pricing

Layer	Price
Base engine (annual subscription)	\$120K / year per bank
Compliance persona pack	\$1,800 / seat / year
Quant / Data Scientist pack	\$2,400 / seat / year
Engineer pack	\$1,500 / seat / year
Trader pack	\$2,400 / seat / year
Wealth Advisor pack	\$2,000 / seat / year

Mid-tier bank ACV (50 seats avg): ~\$982K

3-year SOM across 5 mid-tier banks: \$4.9M ARR

Beachhead

First-year analyst at \$5B–\$50B AUM regional wealth-mgmt or boutique IB, first 90 days, on firm-issued MacBook Pro, working with Excel + CRM + research portal.

Target firms in order:

Raymond James · Edward Jones · Stifel · LPL Financial · Houlihan Lokey · Lazard MM · William Blair · Jefferies · Alvarez & Marsal · FTI · regional banks \$5B–\$50B AUM

Explicitly not Goldman / JPM / Citi — they will build it internally (60%+ AI adoption already).

Risk + escape

Biggest risk: JPMorgan / Goldman / Citi extend their internal LLM Suites to do role-specific scaffolding themselves in 2026–2027.

Escape hatch: Never sell into Top 10. Land at mid-tier where no engineering bench exists.

Secondary moat: bilingual EN / 中文 ship discipline opens China-side mid-tier banks (CITIC / Haitong / HuaTai analyst classes) as secondary market.

Procurement evidence (all public, all reproducible)

Asset	URL
Source code (MIT)	github.com/alex-jb/shadow-mentor
v1.0.0 GitHub Release + benchmark tarball	/releases/tag/v1.0.0
Benchmark history (pre-BR + post-BR + SUMMARY)	/benchmark/history/
8 endpoints API reference	/docs/api-reference.md
Bilingual README (EN + 中文)	/README.md + /README.zh-CN.md
CONTRIBUTING + SECURITY + COC	repo root
llms.txt + robots.txt	repo root
14 consecutive green CI commits	GitHub Actions

Next 90 days

- **Q3 2026** native macOS POC (ScreenCaptureKit + on-device Phi-4-mini)
- **July 2026** 30-target cold email round to mid-tier firms
- **August 2026** Yeshiva Dean / Vice-Provost executive demo via XReal + Flow Immersive
- **August 2026** SOC 2 Type 1 readiness checklist
- **Q4 2026** First paid pilot (\$120K engine + 50-seat compliance pack)

Ask

For the committee / procurement / VC reviewer:

1. Audit the benchmark yourself — `node benchmark/runner.js` against your `ANTHROPIC_API_KEY`
2. Try the MCP server in Claude Desktop — 5-tool surface, no setup
3. Compare honest variance reporting vs other AI banking products' single-run claims
4. Meet for the mid-July XReal + Flow demo

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